THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 9 & 10: **Cell: The Building Block of Life, Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The basic structural and functional unit of life is the: 1

(a) tissue **(b)** organ **(c)** nucleus **(d)** cell

2. If 25 cells lie along a 5 mm diameter of the field of view, the size of one cell is: 1

(a) 0.02 mm **(b)** 2 mm **(c)** 0.2 mm (200 µm) **(d)** 0.5 mm

3. When placed in plain water, a potato piece: 1

(a) stays unchanged **(b)** increases in weight

(c) decreases in weight **(d)** burns

4. Cell wall is absent in: 1

(a) plant cells **(b)** bacterial cells **(c)** fungal cells **(d)** animal cells

5. Prokaryotic cells differ from eukaryotic cells in not having: 1

(a) DNA **(b)** a well-defined nucleus

(c) cytoplasm **(d)** cell membrane

6. Increase in the girth of a dicot stem is due to: 1

(a) intercalary meristem **(b)** lateral meristem

(c) parenchyma **(d)** apical meristem

7. The process by which meristematic cells become specialised is: 1

(a) osmosis **(b)** differentiation **(c)** photosynthesis **(d)** transpiration

8. Root hairs mainly: 1

(a) store food **(b)** provide support

(c) perform photosynthesis **(d)** increase the surface area for absorption

9. The tissue made of dead cells with lignin-thickened walls is: 1

(a) collenchyma **(b)** parenchyma

(c) sclerenchyma **(d)** phloem parenchyma

10. Tracheids and vessels are the conducting elements of: 1

(a) bone **(b)** epithelium **(c)** phloem **(d)** xylem

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why are mitochondria called the powerhouses of the cell? 2

12. State the function of lysosomes. 2

13. Name the three types of muscle tissue. 2

14. Differentiate between a tendon and a ligament. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What is cell culture? State two uses. 3

16. Why are cardiac muscles specialised? How do they differ from skeletal muscles? 3

17. Explain programmed cell death and its importance. 3

OR

What are cancer cells? How do they differ from normal cells? 3

18. What is the role of nervous tissue? Name its structural unit. 3

OR

Name four types of joints with an example of each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** In cell culture, cells are grown in a nutrient medium at a fixed temperature. 5

(i) What is cell culture?
(ii) Why must conditions be sterile?
(iii) Name one use.
(iv) What is totipotency?

OR

(i) Explain programmed cell death. 5
(ii) How do cancer cells differ from normal cells?
(iii) Why is cell division regulated?

20. (i) What is a joint? 5
(ii) Describe four types of joints with an example each.
(iii) How do ligaments and tendons help?

OR

Case study: A yoga practitioner bends the spine and breathes deeply. 5
(i) Which part is flexible due to discs?
(ii) What are discs made of?
(iii) How does the rib cage change in breathing?
(iv) State two health benefits of yoga.

END OF QUESTION PAPER — Total: 40 Marks

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